EXP NO: 2a SUPPORT VECTOR MACHINE (SVM) AND RANDOM FOREST FOR BINARY & MULTICLASS CLASSIFICATION

AIM

To build classification models using Support Vector Machines (SVM) and Random Forest, apply them to a dataset, and evaluate the models using performance metrics like accuracy and confusion matrix.

ALGORITHM

Part A: SVM Model

1. Import necessary libraries
2. Load and explore the dataset
3. Handle missing values if any
4. Encode categorical variables
5. Split dataset into training and testing sets
6. Build SVM classifier using SVC()
7. Train and predict
8. Evaluate the model using accuracy and confusion matrix

Part B: Random Forest Model

1. Initialize Random Forest using RandomForestClassifier()
2. Train and predict
3. Evaluate and compare with SVM

CODE:

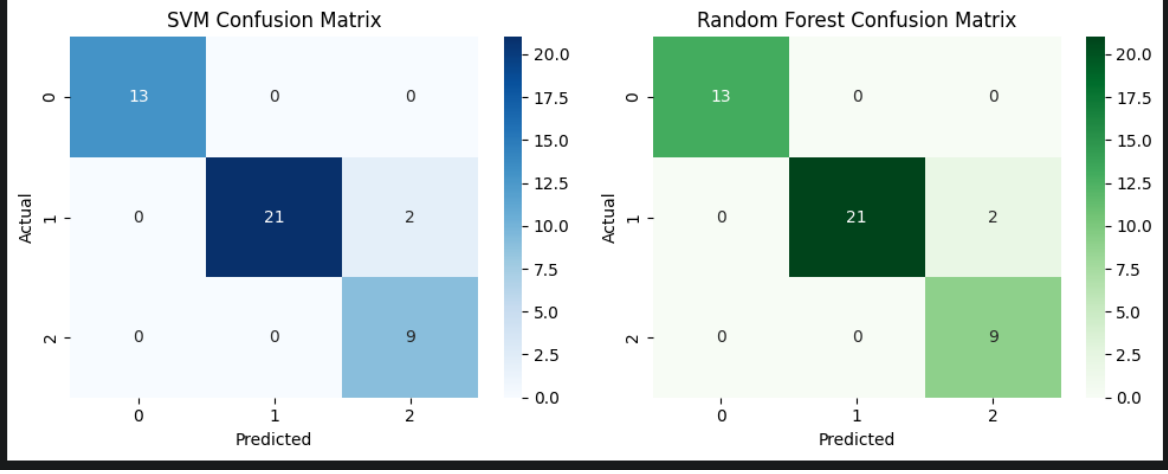
```
import pandas as pd
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
wine = load_wine()
X = wine.data
y = wine.target
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
   X_scaled, y, test_size=0.25, random_state=10
)
svm_model = SVC(kernel='rbf')
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))
rf_model = RandomForestClassifier(
   n_estimators=150,
   random_state=10
)
rf_model.fit(X_train, y_train)
```

```
y_pred_rf = rf_model.predict(X_test)
print("\nRandom Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Random Forest Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.heatmap(
   confusion_matrix(y_test, y_pred_svm),
   annot=True,
   cmap="Blues",
   fmt="d"
)
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.subplot(1,2,2)
sns.heatmap(
   confusion_matrix(y_test, y_pred_rf),
   annot=True,
   cmap="Greens",
   fmt="d"
)
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()
```

OUTPUT:

```
SVM Accuracy: 0.9555555555555556
SVM Confusion Matrix:
[[13  0  0]
 [ 0 21  2]
 [ 0  0  9]]

Random Forest Accuracy: 0.9555555555555556
Random Forest Confusion Matrix:
[[13  0  0]
 [ 0 21  2]
 [ 0  0  9]]
```



RESULT:

The Support Vector Machine (SVM) and Random Forest algorithms were successfully implemented for both binary and multiclass classification tasks. The models were trained and tested on the given dataset, and both achieved good accuracy.